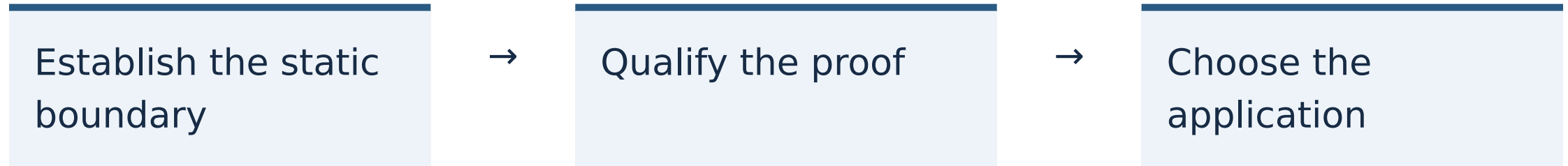


Static and formal verification

Every technique must discharge a named verification claim.



Learning outcomes

Demonstrate these decisions on a real plan row.

- Separate structural results from protocol obligations.
- Qualify a proof by assumptions and abstraction.
- Distinguish generated intent from independent checking.
- Combine typed results through a verification plan.

Route through the lesson

Four blocks, each ending with a decision.

- Establish the static boundary
- Qualify the proof
- Choose the application
- Combine evidence without flattening it

Static results have a subject

Structural cleanliness does not establish functional correctness.



Synchronizers do not define a protocol

Independent bit crossings do not guarantee a coherent word.

- Review the crossing structure.
- Then review data stability, capture and handshake behavior.

Reset and unknowns need their own review

A clock-domain inventory does not enumerate reset hazards.

- Inspect reset assertion and release interactions.
- Separate X optimism from X pessimism.

Discussion: a clean crossing report

Name the evidence that would settle the decision.

- A multibit status uses independent synchronizers.
- The structural report is clean.
- What coherent-transfer claim still needs evidence?

A proof verdict is a qualified statement

Property, model and assumptions travel with the result.



A bounded pass has a boundary

State the depth and justify what lies within it.

- A counterexample is a concrete failing trace.
- A bounded pass needs a depth argument.

Abstraction changes the trace set

Know which conclusion survives the reduction.

- Adding behavior can preserve proofs but introduce false failures.
- Removing behavior can preserve failures but hide real ones.

Discussion: assume away the failure

Name the evidence that would settle the decision.

- An ordering property proves after adding an assumption.
- No owner can justify that environment restriction.
- Can the row close?

Automation still needs reviewed intent

Generated properties inherit their source description.

- Connectivity and register checks scale repetitive obligations.
- Review the description that generated the checks.

An inferred property is not an independent oracle

Mining a design can reproduce its mistake.

- Label the origin of generated properties.
- Compare design-derived claims with external intent.

Equivalence inherits the reference

Preservation and correctness are different claims.

- Define the correspondence and transformation boundary.
- Choose sequential reasoning when state alignment changes.

Discussion: generated from one source

Name the evidence that would settle the decision.

- The register RTL and its checks share a description.
- All generated checks pass.
- Which error can both artifacts share?

Allocate engines per claim

A feature name alone does not select a method.

- Match the oracle, state space and runtime need.
- Keep each result attached to its plan row.

Check the formal environment in simulation

An assumption is a promise that another boundary must honor.

- Turn relevant premises into simulation checks.
- Cover important boundary behavior.

Keep coverage evidence typed

Simulation activity and formal coverage have different denominators.

- Unify the plan, not unrelated percentages.
- Handoff assumptions, exclusions and unresolved traces.

Discussion: the blended dashboard

Name the evidence that would settle the decision.

- A dashboard averages simulation and formal coverage.
- The resulting percentage rises.
- Which closure argument is still missing?

Exercise: coverage or mutation sensitivity?

Choose an instrument for the actual question.

- Proof-core coverage is offered as arbitration-change sensitivity.
- Explain the gap and choose the appropriate instrument.
- Include the chapter's recommended proof-core scope.

What to change on Monday

Make one existing verification decision reviewable.

- Audit an assumption without a discharge owner.
- Find a crossing with no protocol checker.
- Replace one blended metric with per-claim evidence.

Scope and next reading

Use the handout for the detailed artifacts and sources.

- Read the Static and formal verification handout and book Chapters 13–16.
- Tool recipes and complete proof harnesses remain in the handout.